

Standard Form & Indices

Test 4 · Answer sheet · 41 marks total

Section A — Laws of indices

A1. 5^9 (1)

A2. 9^6 (1)

A3. 7^8 (1)

A4. $4^2 \times 4^1 = 4^3$ (1) = **64** (1)

Section B — Zero and negative indices

B1. **1** (1)

B2. $1 \div 2^3 = 1 \div 8$ (1) = **1/8** (1)

B3. $1 \div 10^6 = 1 \div 1000000$ (1) = **0.000001** (1)

Section C — Writing standard form

C1. 9.1 (1) = **9.1×10^4** (1)

C2. 5.6 (1) = **5.6×10^6** (1)

C3. 3.9 (1) = **3.9×10^{-3}** (1)

C4. 8 (1) = **8×10^{-5}** (1)

Section D — Converting back, and calculating

D1. **34,000** (1)

D2. **0.00009** (1)

D3. $6 \times 5 = 30$, $10^3 \times 10^4 = 10^7$ (1) = 30×10^7 (1) adjust to standard form: **3×10^8** (1)

D4. $8 \div 4 = 2$, $10^9 \div 10^4 = 10^5$ (1) = 2×10^5 (1) already in standard form: **2×10^5** (1)

Section E — Powers to know, and estimating roots

E1. **125** (1)

E2. $11^2=121$, $12^2=144$ = between **11 and 12** (1)

Section F — Laws of indices with algebra

F1. u^{13} (1)

F2. v^6 (1)

F3. w^{10} (1)

F4. $6 \times 2 = 12$ (1) $f^{3+5} = f^8$ (1) = **$12f^8$**

F5. $3^3 = 27$ (1) $g^{2 \times 3} = g^6$ (1) = **$27g^6$**

Section G — Fractional indices (Higher)

G1. $\sqrt{64} = \mathbf{8}$ (1)

G2. cube root of 125 = **5** (1)

G3. cube root of 27 = 3 (1) $3^2 = \mathbf{9}$ (1)

G4. $\sqrt{25} = 5$ (1) $1 \div 5 = \mathbf{1/5}$ (1)

Marking note: award method marks for a correctly identified front number/power, or correctly applied law of indices, even if the final answer contains an arithmetic error. Accept equivalent fractions/decimals.